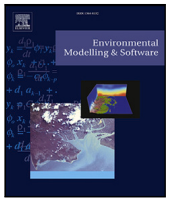
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Position Paper



A novel conceptual flood inundation model for large scale data-scarce regions

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ABSTRACT

We propose a novel conceptual flood inundation model that can be coupled with any hydrological model to generate the probabilistic flood extent maps for a given time step. We generate the runoff rasters using the dynamic Budyko hydrological model that uses the regional IMD/CPC precipitation datasets and the global JAXA GSMaP gauge-precipitation (V7) product over Kerala/Louisiana flood events during August 2018/2016. A freely available global dataset is used for obtaining river geometry information and topographic information is obtained from ASTER/SRTM DEM and localised LiDAR DEM. We evaluated the proposed model using the ground observed flood points from the state disaster management agency for Kerala and flood maps published by USGS for Louisiana. The proposed model captured ~ 45% of flooding during peak flood days while taking ~ 15 min for each run on a daily time step on a desktop computer.

1. Introduction

With the unprecedented availability of space-borne satellite observations, it is now possible to operationally monitor floods using a variety of monitoring approaches, including optical, radar, and the recent GNSS-R satellites, which allow the mapping of flood extents (Yan et al., 2015), depths (Fayne et al., 2017) and in some cases, flow velocity as well (Gleason and Durand, 2020). However, even though these observations help characterise the flood event as it unfolds in near real-time and are useful for hindcast modelling (Bates, 2012, Aristizabal et al., 2020, Unnithan et al., 2020), they cannot be directly used to forecast flood extents (Jain et al., 2018). Instead, flood inundation models (Mishra et al., 2022) are used for predicting the spatial extents and depths of a flood event of a given rainfall event. Different scenarios in which these flood inundation models are used generally consider the fluvial, pluvial, flash, or coastal nature of the flood event. Fluvial floods (Wheater, 2002) are associated with inundation occurring due to overbank flow from the river channel or nearby water body boundary. In contrast, pluvial floods can be attributed to the poor drainage capacity of the soil as observed at the foothills of elevated regions (Kundzewicz et al., 2017) or due to the overflow of drainage systems in an urban setting (Li and Willems, 2020). Flash floods (Blöschl et al., 2008) can be caused by a sudden occurrence of intense precipitation, dam breakage, or unforeseen release of water by control structures upstream along a river channel. Tides and storm surges commonly cause flooding along the coastal regions (Malakar et al., 2021). Each scenario is characterised by different physical processes that require different types of data on the ground for modelling the inundation extents, mainly rainfall gauge stations and discharge monitoring installations that provide frequent observations and are adequate in number across the region of interest. In most instances, these requirements are not met for the inundation model to reliably estimate the inundation extent. Hence the reliance on meteorological models for gridded rainfall data input. In addition, high-resolution topographic information of the landscape in the form of a fine-resolution Digital Elevation Model (DEM) is also needed for flood relief and useful mitigation measures to be put in place (Muhadi et al., 2020).

With 113 million people exposed to floods every year, India is one of the world's most flood-prone countries (Luo et al., 2015). According to the UN (CRED, 2017), India's average annual economic loss due to disasters is estimated to exceed $9.8 billion, out of which more than $7 billion are due to floods. The inundation as a consequence of the intense monsoonal rainfall events is the most common cause of flooding in India (Dhar and Nandargi, 2003). The flood events generally last for about a week, where runoff/discharge from intense precipitation depends upon the ground surface characteristics, including antecedent soil moisture and mean surface elevation (Woods and Sivapalan, 1999). These floods bring about a geomorphic change in the channel floors and increase flood hazards in downstream river reaches. Here, flood hazard

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can be described as the probability that a given spatial region gets inundated in a given time period. The inundation models can thus be further used for flood risk assessment by combining flood hazard information with exposure of humans, livestock and the environment prone to the spatial extent of flood and the vulnerability of each to the flood hazard. However, estimating flood hazards in itself is a time-consuming process because of the inherent complexity in modelling inundation dynamics in a 3D space. In addition, low-resolution DEMs can lead to erroneously small flood extent mapping that cannot be practically used in flood warnings. However, the finer the DEM resolution, the higher the computational complexity and cost in resolving the extent of inundation in the floodplain region. Thus, in data-scarce countries like India where discharge observations from most of the catchments are poor or unavailable, conceptual hydrological and hydraulic models are crucial in helping policymakers in directing relief and adopting precautionary measures.

Traditional flood models are empirically-based and help predict streamflow data provided observed rainfall and discharge time-series data are available. Even with the advent of satellite-based rainfall measurements and the increase in computational power of modern computers, contemporary hydrological models have facilitated the understanding of the complex hydrological process at the expense of inherent model equifinality (Beven, 2001). This implies that complex hydrological models do not guarantee better performance than simpler hydrologic models (Orth et al., 2015). Also, hydrological models do not provide the spatial extent of the spread of excess discharge observed during floods. On the other hand, a simple, calibration-free continuous Rainfall-Runoff (RR) model can be used for driving downstream applications, including hydrodynamic/inundation modelling to predict flood extents. Such RR models will not be affected by the quality or quantity of gauged discharge data available on the ground for calibration, but rather be dependent on the quality of precipitation data available, and on inherent assumptions made on spatial variation of rainfall.

Flood inundation models are often based on a solution of the Saint-Venant hydrodynamic flow equations (Yu and Lane, 2006, Shastry et al., 2019, Sridharan et al., 2020) in 1D, 2D, or 3D, depending on the availability of computational power and topographic information. For locations where channel geometry and highly resolved elevation information have been determined and are available, the Hydrologic Engineering Centre – River Analysis System (HEC-RAS) is a popular standalone hydraulic model that can solve uniform flow or unsteady flow in 1D (Brunner, 2002). Simplified forms of the Saint-Venant equations, namely the diffusive wave and kinematic wave equations that are data-intensive in nature, are employed by large-scale river models, including Catchment based Macro-scale Floodplain (CaMa-Flood) (Yamazaki et al., 2011) and LISFLOOD (Bates and De Roo, 2000). Muskingum routing (Barry and Bajracharya, 1995) and Unit-Hydrographs (Biswal and Singh, 2017) are conceptual, non-dynamic approaches that simulate the flood wave movement and attenuation but do not provide information on flood velocity and inundation depth. The hydrograph-based methods can consistently map the low to medium-range discharge time series, while peak discharges associated with flood events are mostly missed. Recently, there has been an emphasis on the conceptual representation of flow processes (Teng et al., 2017, Zhou et al., 2021), evident in the TVD model (Teng et al., 2015), where inundation extents are pre-processed for a given discharge level in a river reach and extents are identified based on discharge computed from a river system model. Other conceptual techniques include the Rapid Flow Spread Method (Lhomme et al., 2008, Bernini and Franchini, 2013) which could simulate the inundation extents across several basins in the UK in quick time, the Auto-Route Model (Follum et al., 2017) which is an extension of the Muskingum-based discharge routing mechanism that has been tested for a heavy rainfall event that occurred in 2011 across the Lower Mississippi region, and the Cellular Automata (CA) - based flood methods (Jamali et al., 2019), wherein simplified transition rules are employed between raster-based buckets, instead of the shallow water equations. These CA methods suffer from extensive computational time overhead when the higher resolution of the raster buckets is considered for simulating inundation across large geographical regions (Guidolin et al., 2016). In addition, most of the conceptual flood inundation models involve the calibration of parameters that depend upon the reliable estimation of discharge from the macro-scale hydrological models (Venkatesh et al., 2018) or passive satellite observations (Kim and Sharma, 2019) and upon the availability of river geometry data.

Here, a new inundation model is proposed that can be integrated into any hydrological model for mapping inundation at large-scale and for data-scarce regions at near-real-time without the heavy computational overhead. This is demonstrated with the Dynamic Budyko (DB) hydrological model (Biswal, 2016), a calibration-free, continuous rainfall-runoff model as input for the routing module. Discharge in river channels is empirically derived from Manning's equation (Manning, 1891) and conceptually routed across the connected reaches through to the river mouth. Pre-processed extents corresponding to Height Above Nearest Drainage (HAND) (Nobre et al., 2011) values within a given Reach Contributing Area (RCA) are mapped to the conceptually derived discharge values for every reach. These pre-processed HAND inundation extents and corresponding flood depths for each RCA are stitched together to obtain usable flood maps associated with an extreme rainfall event. Finally, an uncertainty analysis based on the river geometry data is carried out, resulting in a probabilistic inundation map for a flood event.

2. Data

We evaluate model performance between JAXA's Global Satellite Mapping of Precipitation (GSMaP) gauge-adjusted (V7) reanalysis product (Kubota et al., 2020) for both Kerala (India) and Louisiana (USA), the Indian Meteorological Department's (IMD) 0.25° gridded precipitation dataset (Pai et al., 2014) over Kerala, and the CPC 0.25° gridded precipitation dataset (Chen and Xie, 2008) over Louisiana. The precipitation and corresponding Land Surface Temperature (LST) products were used to force the DB hydrological model during the August 2018 flood event that led to significant damage in the Indian state of Kerala and for the severe flooding that was occurred across Louisiana in August 2016. Those dates were chosen because both floods were brought about by short, yet highly intense rainfall events. The catastrophic impact resulted in more than 400 fatalities in Kerala (KSDMA, 2018) and economic loss in the order of US$10 billion in Louisiana (Meyer et al., 2020).

Currently, globally available open-access DEMs have a spatial resolution of around ~30 m and a comparable vertical resolution of ~1 m. NASA's Shuttle Radar Topography Mission (SRTM) (Farr et al., 2007) and Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) Version 3 (GDEM 003) (NASA et al., 2019) provide 30 m resolution near-global open-access DEMs, and the model performance was compared across different DEM resolutions over Kerala and Louisiana. The proposed model was also executed in the Louisiana flood event for the high-resolution 10 m LiDAR DEM (Archuleta et al., 2017). Observed flood inundation extents were used to evaluate model performance, using a set of 12,000 flood points from the Kerala State Disaster Management Agency (KSDMA, 2018) for the August 2018 floods, while USGS flood maps were obtained from Watson et al. (2017) for the Louisiana flood event. Empirically derived river geometry data was accessed from Andreadis et al. (2013), which considers bankfull river channel depth and width to follow geomorphic relationship (Leopold and Maddock, 1953) and calibrated with observed discharge values for rivers worldwide. The proposed model was realised with Python 2.7, using the ArcPy package running on top of ArcGIS 10.7 software with ArcHydro plugin added.

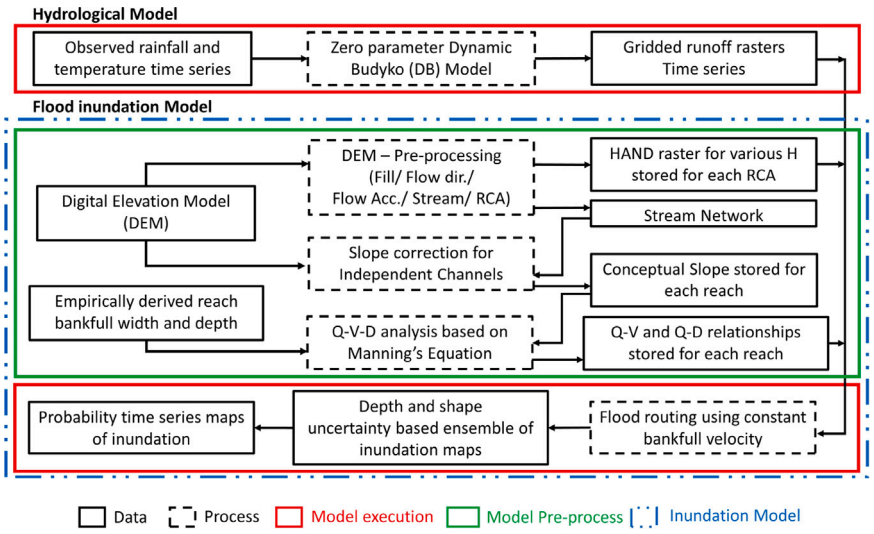


Fig. 1. Overview of the proposed model.

3. Methodology

The proposed model along with sub-components are represented in Fig. 1. Precipitation time-series input is transformed into Effective - Rainfall (ER) time series using the DB model. The DB model considers a calibration-free instantaneous dryness index $\phi(t)$ that performs continuous hydrologic partition at a fine temporal scale. It employs a modified Budyko function $f(\phi(t))$ that mimics the effect of antecedent rainfall and solar radiation on the ER at a given time instant, according to the following equation,

$$ER(t) = W(t) \cdot (1 - (\phi.tanh(\phi^{-1})) \cdot (1 - e^{\phi^{-1}}))^{0.5} \quad (1)$$

Where $W(t)$ refers to the rainfall water available after satisfying current losses due to evapotranspiration (ET). In Eq. (2), $(\phi(t))$ given by is defined as the ratio of the total functional rainfall (FW) to the total functional solar energy (FH) for the preceding time period n in days, which is taken at 365 (Biswal, 2016) and H denotes the remaining available energy after conceptual flow partitioning framework adopted by Biswal (2016). The FW and FH terms are obtained by applying the empirically derived 'universal' decay function $x(t) = x(0) \cdot (1 + 0.4 \cdot t)^{-1}$ (where $x(t)$ denotes the rate of decay of $x(0)$ at time t) to the past available water and potential evapotranspiration (PET) time series values to replace the soil moisture content. The decay function is assumed to be the same for catchments located across all geographic locations.

$$\phi(t) = \frac{FW}{FH} = \frac{\int_{t-n}^{t} W(\tau) \cdot x(t-\tau) \cdot dt}{\int_{t-n}^{t} H(\tau) \cdot x(t-\tau) \cdot dt} \quad (2)$$

$$Ru(t) = -\frac{d}{dt}\left[\int_{0}^{t} ER(\tau) \cdot x(\phi(t-\tau)) \cdot dt\right] \quad (3)$$

Where, τ denotes a dummy variable. ER decays with time into Runoff(Ru) as a result of the decay function. The negative sign in the raster runoff time series obtained from Eq. (3) indicates rainfall decays into the runoff for the region of interest, which is then given as input to the flood inundation model for near-real-time execution. The distributed runoff time series from any continuous hydrological model can be provided as input to this inundation module. While the proposed model requires only rastered runoff time series for near-real-time execution, much of the topographic and routing modules are pre-processed to save on computational overheads.

For any given region the proposed model can automatically fill the provided DEM at any spatial resolution, derive the flow direction raster based on the D8 algorithm (Jenson and Domingue, 1988), and derive a flow accumulation raster to delineate the drainage/channel network. We consider a Contributing Area (CA) threshold of 9 km^2 for the DEM of 30 m spatial resolution, to reflect upon the computational overhead that needs to be undertaken. All pixels draining into the same reach constitute the Reach Contributing Area (RCA), whose boundary is delineated and the Height Above Nearest Drainage (HAND) raster maps corresponding to varying threshold height values within each RCA boundary are extracted separately to be later mosaiced during model execution to obtain required flood map (refer Fig. 2 shown along with model assumptions). The runoff within the RCA is determined by the area-weighted average method, as shown in Fig. 3. Here, the total runoff Ru_{reach} in an RCA of area A, $(A = A_1 + A_2 + A_3 + A_4)$ that has to be subsequently routed is given by

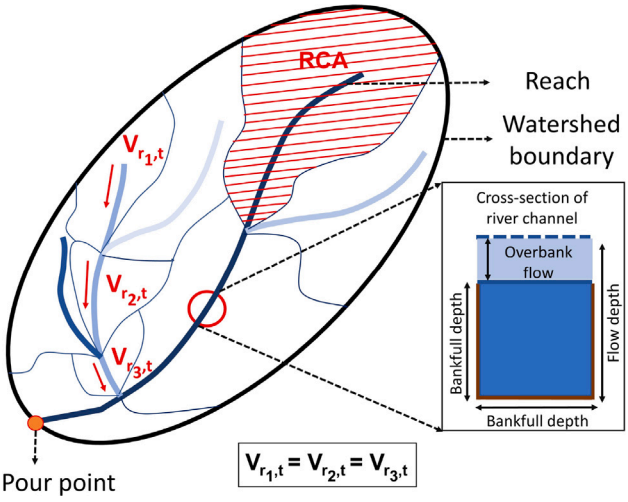


Fig. 2. Reach Contributing Area depicted along with model assumptions.

$$Ru_{reach} = \frac{A_1 Ru_1 + A_2 Ru_2 + A_3 Ru_3 + A_4 Ru_4}{A} \quad (4)$$

The empirical bankfull-width and -depth values were obtained for every reach from Andreadis et al. (2013) and then transferred to the nearest reach segment that we derived from the DEM. This was necessary because the satellite imagery derived river network shapefile developed by Andreadis et al. (2013) does not fully coincide with the rasterized channels delineated from the DEM. Following the pre-processing of all required static river information (i.e. reach length, bankfull-width and -depth), it is now possible to identify the groups of

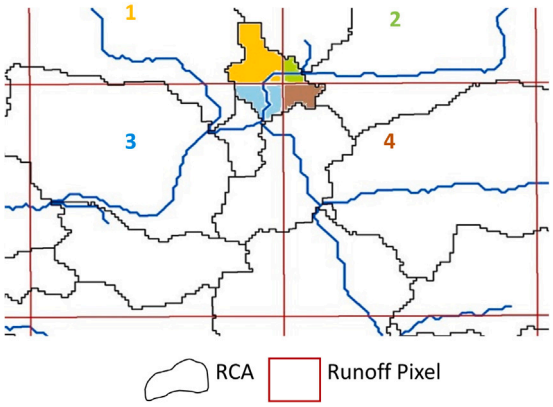


Fig. 3. Area weighted method of estimating discharge within an RCA.

reaches with common pour points belonging to the same river system (see Fig. 2). With Leopold and Maddock (1953)'s observations, the river velocity is assumed to be spatially constant for the family of reaches having a common outlet for a given time interval. The velocity within a channel reach (V_{reach}) can then be determined from Manning's equation as a function of the flow width w and depth d. Assuming a rectangular channel geometry, the hydraulic radius r (i.e. the ratio of the area of the flow's cross-section (a) and its wetted perimeter (p)) can be explicitly calculated and Manning's solved for the flow velocity V_{reach} as follows:

$$V_{reach} = \frac{1}{\eta} \cdot \frac{(w \cdot d)^{2/3}}{(w + 2d)^{2/3}} \cdot S^{1/2} \tag{5}$$

Where η denotes the roughness coefficient of the channel and S is the channel slope. Here, it is assumed that $\eta = 0.03$, in line with what other global routing and inundation models have considered for model execution (Yamazaki et al., 2011). This assumption is a first-order simplification, necessary due to the lack of reliable and extensive spatial roughness data. Velocity in a reach is largely affected by the flow depth and the slope of the channel bed. This is because we can more reliably calibrate river width than the location's depth with satellite imagery data. Here, a slope correction technique is proposed that can be incorporated for each reach in the following manner. In order to achieve this, the channel network is first divided into a set of *independent* channels (Biswal and Marani, 2014) as shown in Fig. 4. An independent channel is therefore defined by a junction and source characterised as its end nodes. The node/pixel where two reaches meet in a channel network is called a junction, while an upstream reach head pixel is called the source node. The river channel can then be visualised with n source nodes and $n-1$ junction nodes, with the river mouth node designated as a pseudo nth junction node. We identify an independent channel as a set of reaches between the nearest source node for a particular junction node. Once a set of independent channels is successfully delineated from the river network, the elevation (e) profile is pixel-wise extracted from the DEM along each independent channel. The distance of each pixel to the downstream mouth (d_{Outlet}) is stored and the relationship between e and d_{Outlet} is modelled via exponential distribution curve fitting. Once the curve parameters have been derived, the conceptual slope (S_c) can be evaluated for every channel pixel in the independent channel by substituting d_{Outlet} for that pixel. Using S_c in Eq. (2), the routed discharge Q_{reach} can be calculated as follows,

$$Q_{reach} = V_{reach} \times A_{cs} \tag{6}$$

Where A_{cs} denotes the cross-sectional area of the channel geometry. For each reach, a set of simulated discharge (q) - velocity (v) pairs needs to be developed for different flow heights (h) considering both over-and-under-bankfull height conditions. In the case of over-bank

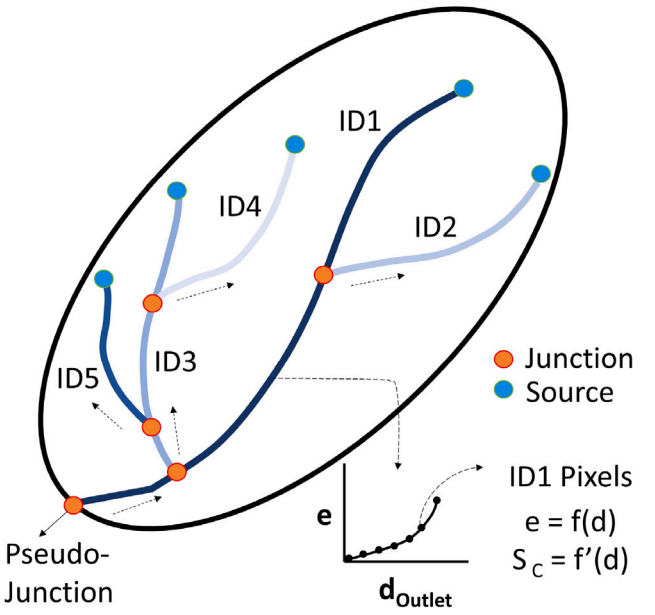


Fig. 4. Characterising river network as a set of Independent Channels (ICs), with pixels along a given IC used for conceptual slope computation.

flow, we assume that the entire q is routed only across the channel. This assumption is made such that over-bank flow is blocked by natural/man-made obstacles and over-bank flow merely inundates. Furthermore, v is simulated considering the wetted channel perimeter as constant at $W + 2D$, where W and D denote the corresponding bankfull width and depth of that particular reach. Correspondingly, q - h values are also stored for each reach based on this pre-processing method.

The bankfull velocity is then assigned to all the reaches in the river system to calculate the flow height (h) from the pre-computed values of q - h for each reach. The corresponding inundation depth within a RCA (D_T) is obtained by reducing the bankfull depth from the h obtained for a reach, given as (refer Fig. 5),

$$D_T = h - D \tag{7}$$

Respective HAND rasters based on the D_T obtained for each RCA are then chosen and mosaiced together to obtain the flood inundation map across the entire region of interest. Finally, an uncertainty analysis was performed to ascertain the error associated with the empirically derived bankfull-depth values and the shape of the river channel by considering a parameter θ (shown in Fig. 6) that denotes the incidence of the channel wall with the river bed. In the case of large-scale flood events where flow discharge (q) is much higher than the bankfull channel carrying capacity, the flow height and ensuing flow depth affect the extent of inundation to a greater extent than changes in η roughness coefficient (Eilander et al., 2023). In addition, the spatial and temporal variability of η is data-intensive and unavailable, especially in data-scarce regions. In such scenarios, the inundation modelling involves significant computational complexity, and η reduces to a model parameter with less physical understanding because of high natural variability.

4. Results and discussion

A flood inundation model is predominantly dependent on the rainfall data driving the hydrological model and the spatial and vertical resolution of the available DEM for producing usable flood maps. Thus, to identify the proposed model's main performance drivers, different precipitation products for the same flooding event were considered

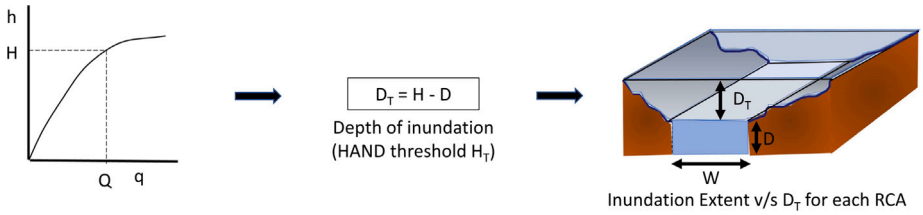


Fig. 5. Conversion of flow height h into inundation depth D_T.

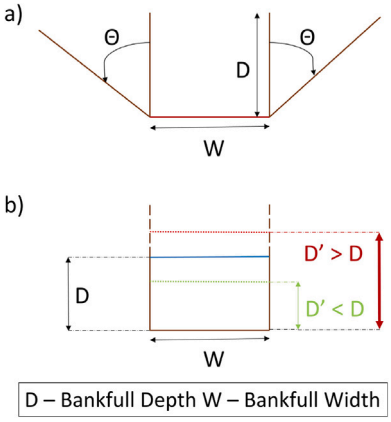


Fig. 6. Uncertainty analysis of proposed model considering (a) Uncertainty in shape of the reach (b) Uncertainty in empirically derived bankfull height values.

across vastly different landscapes at distinct spatial resolutions. The 0.25° IMD and 0.10° GSMaP rainfall rasters for the peak flood day across Kerala (16 August 2018) are shown in Fig. 7a. While the 0.25° CPC and 0.10° GSMaP rainfall rasters capture the peak flood event in Louisiana on 15 August 2016. We see that both events are characterised by heavy rainfall over extensive inland regions. It was found that the regionalised IMD/CPC precipitation products for Kerala and Louisiana capture extreme rainfall events with a higher magnitude than those recorded in the GSMaP product. This, in turn, affects the runoff derived from the DB model corresponding to different precipitation datasets. Higher runoff is observed during the flooding period, subject to IMD/CPC precipitation for Kerala/Louisana, than in the case of GSMaP based runoff from DB, as shown in Fig. 7b.

The runoff has been routed using our proposed method through which the discharge time-series for the Periyar and Pamba river basins in Kerala (indicated in Fig. 7c) is obtained, both of which witnessed significant flooding during August 2018. The peak discharge reaches 3325 m^3/s and 3147 m^3/s on August 16 for the Pamba and Periyar river basins, respectively, using the IMD precipitation product. The discharge time series based on the GSMaP dataset also peaks simultaneously with the IMD rainfall-based discharge; however, the difference in peak discharge can be attributed to the difference in precipitation between IMD and GSMaP. Observed flow rates were available for the Periyar river basin, with a peak value of 5900 m^3/s on 16 August (CWC, 2018). We can thus infer that although the discharge time series gauged by the proposed method exhibits skill in capturing the timing of the inundation, it underestimates the peak flood discharge value. This can be attributed to the contributions of the extremely rare rainfall event with a 100-year return period, the error in rating curve estimations for such extreme values and also due to the dam operations in the state, most of which were opened during the flood event (Mishra and Shah, 2018, Mishra et al., 2018). Furthermore, the DB model output underestimates the quick flow component of the runoff, which although will not affect total flow across the flood period, can decrease the modelled peak discharge. Nevertheless, the discharge time series were evaluated for Kerala, which also seems to follow a similar temporal pattern for both basins. This is due to the daily time step associated with IMD and GSMaP rainfall rasters, the relatively short length of the river network, with Periyar being the longest river in Kerala at ~250 km and similar catchment area characteristics. We observe that there is almost no time lag of more than a day associated with the discharge for all river basins in Kerala.

Model discharge estimated for the river Mermentau and Tickflaw in Louisiana peaked at 310 m^3/s and 347 m^3/s, respectively in August 2016. Upon comparison with the available peak gauged discharge values from USGS, we found that the instantaneous peak of 1145 m^3/s (Mermentau) and 1010 m^3/s (Tickflaw) was observed during August 2016. It is to be noted that observed discharge time series were not available for this particular extreme flood event from USGS across different gauging stations in the state. It is also noted that the instantaneous peak values do not compare well with the overall discharge observed across an entire day. Also, in the case of flood events, the discharge rating curves that are followed at particular gauging stations will cease to hold in case of extreme level-gauge flood heights. We, therefore, draw comparisons directly between model-derived probabilistic inundation extents (Fig. 8) and observed flood maps rather than compare with respective scarce flood discharge time series. The resultant flood discharge and associated extents are also affected by the region's topography under consideration, in addition to the uncertainty in the precipitation dataset. Kerala is characterised by a dominantly undulated and highly vegetated terrain, surrounded by the Western Ghats along its eastern boundary. Louisiana is a mixture of dense forests, shrubs, and herbaceous terrain, with a gradually decreasing elevation meeting the coastline at a fairly even slope. We analysed the flood extents corresponding to different spatial resolutions of the input DEM for the given study regions. For Kerala, we used the ASTER 30 m, ASTER 90 m, SRTM 30 m, and SRTM 90 m DEMs to delineate the river networks and corresponding RCAs (Fig. 8). Highly resolved 10 m LiDAR DEM was available for Louisiana, which was then spatially resolved to 30 m, 90 m, 270 m, and 810 m that were compared with the model performance upon using the different spatially resolved SRTM/ASTER DEMs. To appreciate the contrast in input DEMs (Fig. 10), we analyse distinct differences between the LiDAR resolved 90 m and SRTM 90 m DEM for river Mermentau. From SRTM, we can see that the river channel pixels cannot be differentiated from floodplain pixels and that the channel pixels have higher elevation values than nearby floodplain pixels in upstream regions. Corresponding to different input DEMs, we generate the flood extents within each RCA corresponding to the $q - h$ relationship for each reach and map the pre-processed HAND raster with the obtained flood depth D_T, shown in Fig. 8.

For Kerala, the proposed model flood extents for the peak flood day on August 16 were generated for the IMD and GSMaP rainfall data as shown in Fig. 7d. We compared the peak flood inundation extent with the ~ 12000 observed flood points across the entire flooding period from KSDMA. Sentinel - 1 A derived flood inundation maps were not used for comparison in this scenario since the satellite overpass over Kerala was on August 21, much after the peak flood event had passed, resulting in the capture of inundation only in a few downstream reaches (only 14% in Flood Accuracy (FA), a validation metric that is explained subsequently, on comparison with observed data from KSDMA). In the

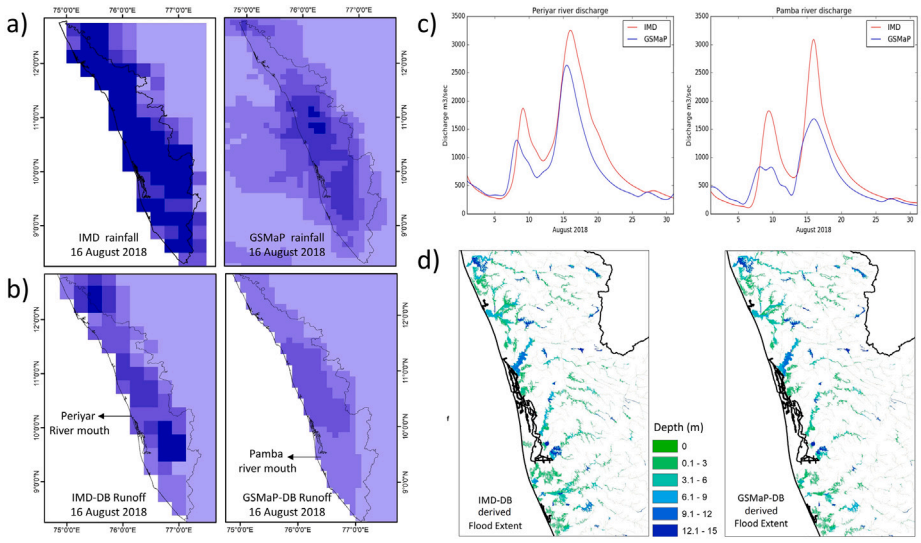


Fig. 7. (a) 0.25° IMD and 0.1° JAXA Precipitation (mm) over Kerala on 16 August 2018 (b) Runoff (mm) obtained for respective precipitation with DB model (c) Discharge time series observed over Periyar and Pamba river basins during August 2018 (d) Flood Extent derived for IMD-DB and GSMaP-DB on 16 August 2018.

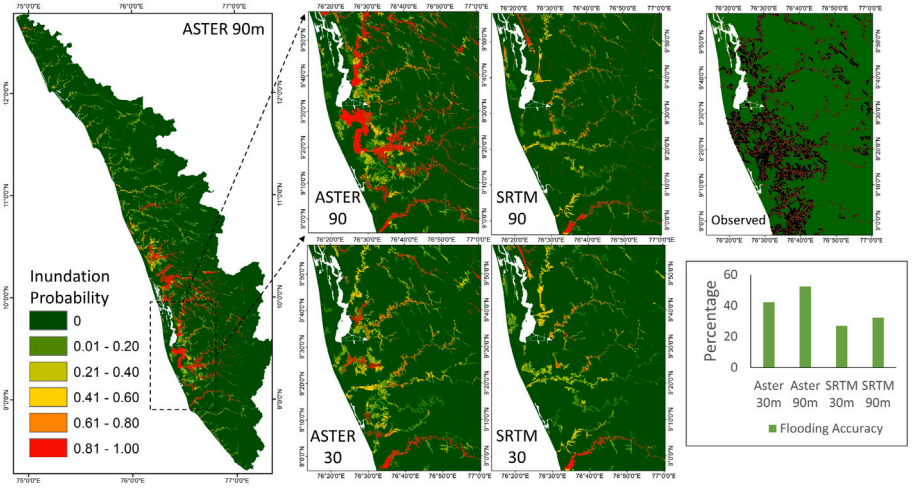


Fig. 8. Model-derived probabilistic flood extents for different spatially resolved input DEMs for the August 2018 flood event witnessed across Kerala, India.

case of the observed flood points from KSDMA, we observed a spatial bias in collecting information on whether a place was inundated or not. Most often, the reported GPS coordinates of the location reportedly inundated followed the access roads when superimposed on Google Earth. We corrected this spatial error by considering all the pixels/points downstream of the reported point to the river channel as inundated. This exercise resulted in a total of ~370,000 pixels being available for comparison with modelled extent. We evaluate the flood extent using Flood Accuracy (FA) indicator, given by

$$FA = \frac{n_F}{n_F + n_{NF}} \times 100 \tag{8}$$

Where n_F denotes the number of pixels captured by the model as flooded, and n_{NF} denotes the number of pixels that failed to be captured by the model as flooded. Here, $n_F + n_{NF}$ is the total number of pixels that we generated earlier for comparison. Uncertainty in the bankfull height values was addressed by sampling between the 95% confidence interval values following Andreadis et al. (2013). The probabilistic mapping of inundation corresponding to the uncertainty in empirically derived bankfull values in the case of IMD/GSMaP input as shown in Fig. 8. Probabilistic mapping of inundation extents corresponding to different values of θ to simulate different channel geometries resulted in similar flood extents. This is because the order of change in hydraulic radius r in Eq. (2) does not achieve a significant change in q to affect the inundation extent. Upon converting the probability inundation maps into binary flood/no-flood maps by considering a pixel to be flooded even if it is associated with a 1% probability of inundation, we observed the highest FA in the case of ASTER 90 m DEM-derived model extent, compared to ASTER 30 m derived flood extent. Similarly, SRTM 90 m DEM-derived flood extent FA outperformed upon comparison with the SRTM 30 m derived flood map. However, upon comparing FA of flood extents derived from ASTER and SRTM 90 m DEMs, we find that ASTER V3 DEM is much more suitable for mapping inundation using the proposed model in the case of mountainous regions like Kerala. The model with IMD and ASTER 90 m DEM input captured a total of 198,112 points resulting in FA of 51%, while GSMaP/90 m ASTER input derived flood extent mapped a FA of 42%. The better performance in model-derived inundation extent can be directly linked to higher precipitation observed by IMD than that observed with GSMaP.

The model performance was further evaluated by comparing it with commercial flood models including TUFLOW. TUFLOW-derived inundation extents worked best in the case of catchment- and reach-specific analysis than in the case of widespread floods witnessed in Kerala. Heavy computational overhead was involved in numerically

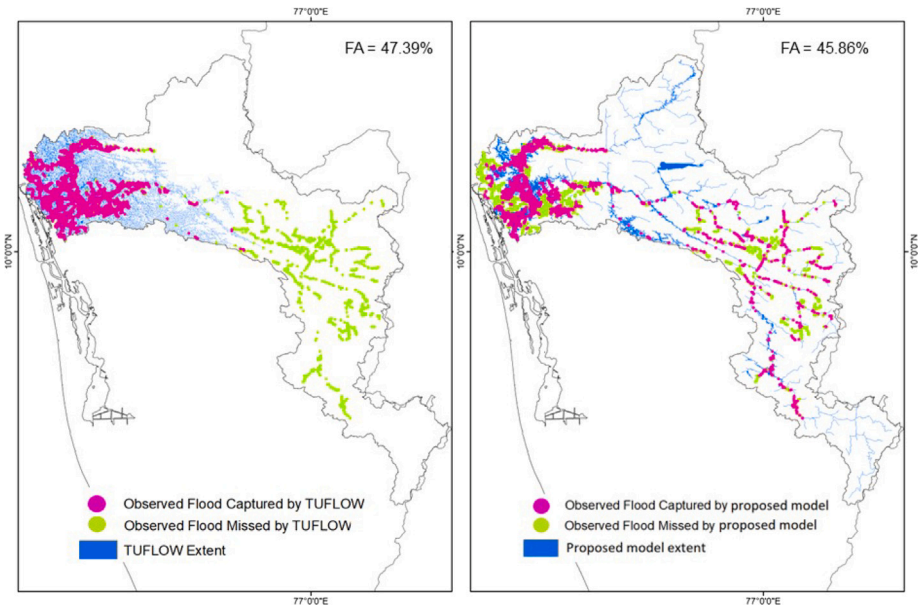


Fig. 9. (a) TUFLOW flood model derived inundation map for Periyar catchment in Kerala compared with observed flood points from KSDMA for August 2018 (b) Proposed flood model derived inundation map for Periyar catchment in Kerala compared with observed flood points from KSDMA for August 2018.

solving the simplified versions of the Navier–Stokes Equation in-built within TUFLOW for the same model parameters ($\eta = 0.03$) and inputs (ASTER DEM at 90 m spatial resolution and Dynamic Budyko gridded runoff time-series) as given to the proposed model. TUFLOW was executed for the Periyar catchment in Kerala on 16 August 2018 with a runtime of 21 h using Intel(R) Xeon(R) Gold 6254 CPU with 3.10 GHz processor as against 8 mins needed for the proposed model on the same computational architecture. An FA of 47.39% was obtained for peak flood extent using TUFLOW, which again compares well with the proposed model-derived FA of 45.86% for the said region (Fig. 9). TUFLOW captures higher observed flood points in the floodplains due to the higher efficiency in solving the 2D Navier–Stokes equations while the proposed model captures more flood points along upstream reaches because of the inherent assumptions in slope corrections.

USGS interpolated observed high watermarks for the Louisiana flood event to produce inundation maps for flooded regions. Since the observed flood map included both inundated and dry pixels, we compared the modelled extents with observed maps using a more intuitive indicator called the Critical Success Index (CSI) given by,

$$CSI = \frac{n_{M_F,O_F}}{n_{M_F,O_F} + n_{M_{NF},O_{NF}} + n_{M_F,O_{NF}}} \times 100 \tag{9}$$

Where n denotes the count of Model (M) / Observed (O) pixels being Flooded (F) / Non-Flooded (NF). The proposed model inundation extents for different sources of DEMs at different spatial resolutions are shown in Fig. 11. With CPC/GSMaP precipitation input, the output obtained with LiDAR-based DEMs outperformed those that utilised SRTM DEMs at the same spatial resolution. This can be attributed to the high accuracy and precision in LiDAR point clouds that determine terrain elevation compared to the radar-based methods affected by the surface vegetation and man-made structures. For the same spatial resolution, ASTER DEMs derived flood extents perform the worst indicating that SRTM DEMs are a more appropriate choice in the case of the relatively flat topography of Louisiana. In addition, the CSI values associated with the coarser resolution DEMs are better than the model results with finer DEMs, which is largely due to the reduced spatial allocation errors. However, even when we executed the inundation model at the coarsest resolution of 810 m, the CSI values associated with model capacity in capturing floods kept improving with a coarser spatial resolution for the same reason. In order to arrive at an optimum spatial resolution that is to be adopted for model execution, we examined another accuracy metric, the Youden's Index (YI), given by,

$$YI = \left(\frac{n_{M_F,O_F}}{n_{M_F,O_F} + n_{M_F,O_{NF}}} + \frac{n_{M_{NF},O_{NF}}}{n_{M_{NF},O_F} + n_{M_{NF},O_{NF}}} - 1\right) \times 100 \tag{10}$$

which captures the overall accuracy of the inundation model across the Mermentau, Vermillion, Amite, Tickflaw, and Tangipahoa rivers in Louisiana. Due to its formulation, the YI captures both flooded and non-flooded regions when compared with observed maps and therefore represents a more accurate true-positive and true-false representation of the inundated areas and dry pixels within the region of interest. The peak model performance, in this case, is at a spatial resolution of 90 m for the LiDAR DEM when mapping accuracy across different spatial resolutions. This can be attributed to the model approximation in slope computations which have the least uncertainty and is more productive at a coarser spatial scale of around 90 m, while fine-scale mapping will result in poorer accuracy as indicated by YI.

We expect improved model performance at finer resolution with access to an accurate river geometry dataset. It can also be observed that the LiDAR 90 m derived inundation map has higher accuracy in steeper terrain compared to the inundation mapped for the Mermentau River in Louisiana (shown in Fig. 11) with a CSI of 0.8 than in the case of flat terrain flooding observed along the Vermillion river (CSI of 0.36). This is because the 2D inundation dynamics dominate flatlands more than mountainous regions. The high error (over-and under-estimation — refer to Fig. 12a and 12b) in the case of flood maps obtained from ASTER/SRTM DEMs can be attributed to the spatially displaced river networks and associated RCAs that are delineated from those corresponding vertical error-prone open-access DEMs. The resulting inundation patterns are also displaced when compared with the observed inundation patterns from USGS. Since the bankfull discharge is conceptually estimated and because the model does not account for non-linearity in flood wave propagation, the over-bank discharge may not be accurately captured. However, the proposed model taking LiDAR DEM as input performs with comparable run-time execution and accuracy commensurate to that of other data-driven rapid inundation models (Jamali et al., 2018, Jamali et al., 2019), with the advantage of incorporating conceptual flow dynamics.

The model's novelty lies in applying the existing concepts of HAND and Manning's equation by making suitable assumptions, such as neglecting overbank flow contribution and exponential elevation profile

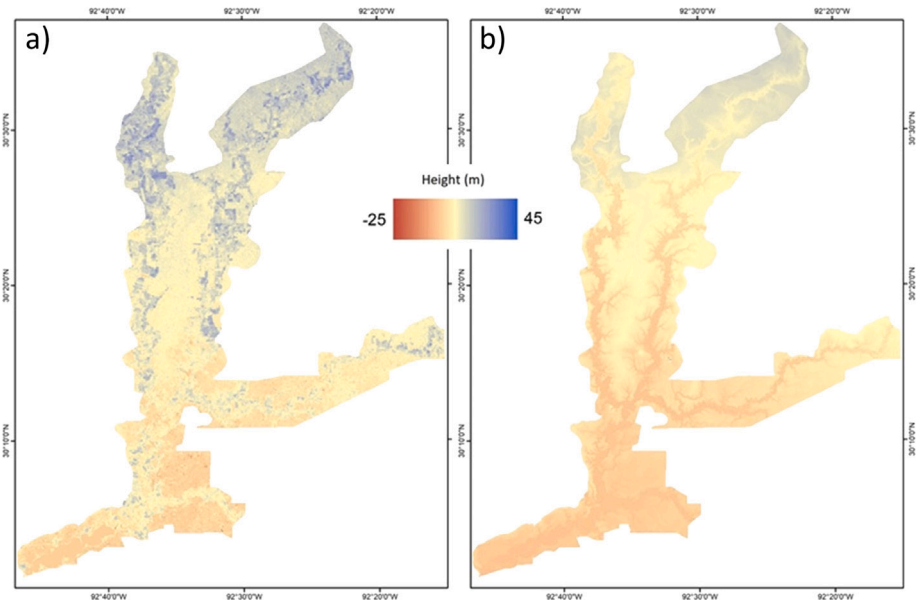


Fig. 10. (a) SRTM 90 m DEM (b) LiDAR-derived resampled 90 m DEM for river Mermentau in Louisiana, USA.

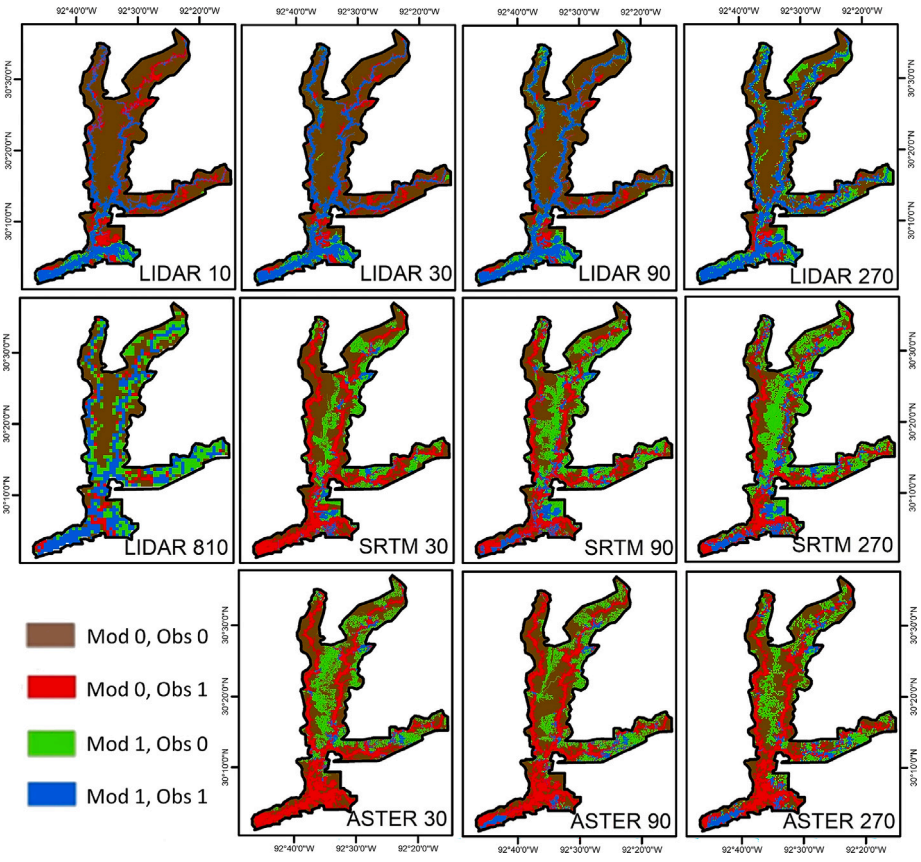


Fig. 11. Model-derived probabilistic flood extents for different spatially resolved input DEMs for the August 2016 flood event for River Mermentau, Louisiana.

assumption for slope estimation. Moreover, several computational challenges were addressed to apply the model, such as merging the channel network from Andreadis et al. (2013) dataset with the DEM-derived channel network. The conceptual inundation model was implemented as a set of Python routines, envisioned to be computationally efficient by preprocessing the static DEM dataset to derive the RCAs, HAND extents corresponding to each RCA, and the flow discharge (q) – velocity (v) – height (h) relationships were pre-computed for different flow conditions for each reach. Upon availability of the time series of the rainfall dataset, the inundation module takes the preprocessed inputs to map flooding at the exact temporal resolution as that of the rainfall dataset. Considering the spatial scale of the flooding witnessed across Kerala (38,000 km^2) and Louisiana (100,000 km^2), the proposed methodology is unique for near real-time execution of time-series rainfall datasets. The inundation model executes sequentially for every time step output (daily in case of DB model output) based on

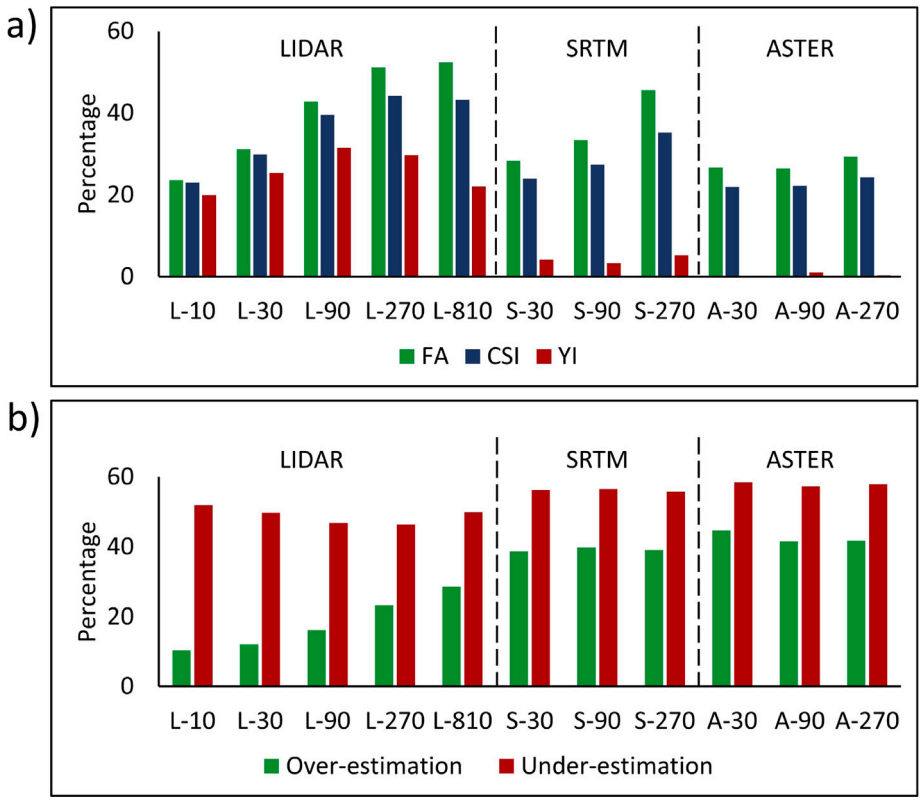


Fig. 12. Model-derived flood inundation map (a) accuracy and (b) error across different sources of DEM's at different spatial resolutions for inundation observed across Mermentau, Vermillion, Tickflaw, Amite and Comite rivers in Louisiana during August 2016.

the hydrological model gridded runoff and involves a 4-step processing pipeline – (a) Area weighted runoff calculation within each RCA (b) Routing runoff and estimating HAND threshold within each catchment system (c) Stitching HAND rasters for all RCAs within study region resulting in flood maps associated with derived flood height and (d) Generation of uncertainty maps for calculating inundation probability. For Kerala, runtimes encompassing all the above steps lasted 15 mins in case of 90 m DEM for single-day evaluation and 27 mins for 30 m DEM. For Louisiana, the model took each observation day 18 min, 29 min, 46 min, 80 min, and 97 min for execution with input DEMs of spatial resolution 810 m, 270 m, 90 m, 30 m and 10 m DEM respectively. In contrast, other commercial flood software like TUFLOW took several hours of execution for a single basin within the study region with comparable flooding accuracy, while global inundation models like LIS-FLOOD produce inundation maps corresponding to stochastic event-based rainfall estimates. CaMa-FLOOD executes at a much coarser scale (with the highest resolution recommended at 1 km) even if it takes the same ASTER and SRTM DEM as input.

5. Conclusion

A newly developed, simplified, fast processing, conceptual flood routing scheme and a large-scale inundation modelling algorithm for data-scarce regions have been presented. The proposed model was evaluated against two flood events in Louisiana (August 2016) and Kerala (August 2018). These flood events were characterised by heavy rainfall across substantial inland regions with different topographic relief and land cover patterns. Regionalised rainfall products like IMD, CPC and the GSMaP were used as input into the Dynamic Budyko hydrological model to determine the impact of rainfall product differences on modelled flood extents. The model performance was also evaluated based on different DEM sources and the input DEMs' spatial resolution for Kerala and Louisiana.

Overall, it was found that the modelling scheme showed good skill in capturing the timing of the flood event. The flood extents were sensitive to the magnitude of precipitation values, as observed with higher FA of IMD-derived flood maps due to higher rainfall intensity captured by IMD than GSMaP for the August 2018 Kerala flood event. The proposed model showed better YI performance at a coarser resolution of 90 m in the case of LiDAR-based DEMs for the August 2016 Louisiana flood event which can be attributed to the model formulation of slope computation. The study thus shows the potential application of the empirically derived Manning's equation in effectively capturing flood extents in large data-scarce regions without using observed discharge time-series information for calibrating flood extents. Rainfall is considered the primary input driving the inundation model apart from topography. The intensity (magnitude) of different time-series rainfall products was found to be the main cause affecting inundation extents across Kerala and Louisiana during August 2018 and 2016 respectively. The approach proposed in the current work is model agnostic and a concept study for modelling inundation with the availability of any hydrological model-based time series gridded runoff dataset. Different hydrological model time series runoffs obtained for the same gridded rainfall product for prolonged or heavy/large rainfall events do not result in large changes in peak flood extent over the same region.

The current study uses bankfull geometry estimates of global river reaches, whereas bankfull depth estimates have been validated only for rivers in the US, and not for data-scarce regions in India. Thus, improvement in local estimates of bankfull geometry can be made possible with regular time series of discharge estimates for individual reaches within the study region of interest. In addition to bankfull depth, inundation is primarily affected by the roughness of the channel bed surface. Uncertainty analysis based on Manning's roughness coefficient can result in a more in-depth understanding of downstream discharge variation with η. Also, HAND-based methods tend to overestimate

inundation in flat deltaic regions, and thus, there is a crucial need to incorporate the branching of the river channel network at the mouth of catchment pour points to capture realistic inundation patterns. In addition, the stagnated waters in floodplain regions can lead to the buildup of inundation extents in upstream regions, called the backwater effect, which needs to be conceptually represented to capture a wide range of inundation scenarios. Nevertheless, the proposed model can be coupled with forecast rainfall values to predict an ensemble of inundation maps at fine resolution and large spatio-temporal scales to help policymakers and emergency managers identify vulnerable places from impending floods in data-scarce regions.

CRediT authorship contribution statement

S.L. Kesav Unnithan: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Basudev Biswal:** Conceptualization, Methodology, Formal analysis, Resources, Writing – review & editing, Project administration. **Christoph Rüdiger:** Investigation, Validation, Visualization, Writing – review & editing. **Amit Kumar Dubey:** Formal analysis, Investigation, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: S.L. Kesav Unnithan has patent #202321007807 pending to Indian Patent Office. Basudev Biswal has patent #202321007807 pending to Indian Patent Office. Christoph Rüdiger has patent #202321007807 pending to Indian Patent Office.

Data availability

The complete dataset will be made available on request.

Sample model code and dataset (CFRHIM Repository).

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